

NOTTINGHAM EAST MIDLANDS, United Kingdom

WMO: 034185

Lat: 52.831N

Lon: 1.328W

Elev: 306

StdP: 14.53

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.4	(c) 28.4	(d) 21.3	(e) 16.2	(f) 30.8	(g) 23.4	(h) 17.9	(i) 31.7	(j) 34.5	(k) 47.7	(l) 31.5	(m) 47.6	(n) 6.9	(o) 270	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	80.2	65.0	75.5	62.9	73.0	61.8	66.6	76.2	64.7	72.8	63.2	70.5	9.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.8	69.8	61.1	81.5	67.8	60.4	79.2	66.9	31.2	76.2	29.8	73.0	28.7	70.2	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	24.3	21.4	21.6	86.1	3.3	3.8	19.2	88.9	17.3	91.1	15.4	93.2	13.0	95.9
			21.8	70.7	3.7	3.4	19.2	73.1	17.0	75.2	14.9	77.1	12.2	79.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	40.2	40.8	43.7	48.0	53.3	58.7	62.7	62.3	57.7	51.8	45.0	40.8		
	DBStd	9.70	5.61	5.79	5.38	5.41	5.33	4.77	4.69	4.41	4.35	5.14	5.63	6.45		
	HDD50	1389	306	261	203	99	26	0	0	0	1	39	164	290		
	HDD65	5373	770	678	659	510	364	198	103	109	223	410	599	750		
	CDD50	1563	1	3	9	39	128	261	395	381	232	95	15	4		
	CDD65	73	0	0	0	0	1	9	32	26	4	1	0	0		
	CDH74	537	0	0	0	1	12	86	252	160	23	3	0	0		
	CDH80	96	0	0	0	0	0	13	49	33	1	0	0	0		
(14) Wind	WSAvg	10.6	12.4	12.2	11.6	10.2	10.0	9.5	9.2	9.3	9.6	10.8	10.9	11.7		
(15) Precipitation	PrecAvg	24.5	2.1	1.7	1.8	1.9	2.0	2.3	2.0	2.2	2.0	2.0	2.1	2.2		
	PrecMax	57.6	5.8	6.9	5.6	4.9	14.7	5.6	8.5	5.4	5.0	5.4	4.3	5.6		
	PrecMin	17.6	0.5	0.2	0.3	0.3	0.4	0.3	0.2	0.2	0.3	0.3	0.9	0.4		
	PrecStd	6.5	1.1	1.3	1.0	1.1	2.3	1.4	1.6	1.1	1.2	1.1	0.8	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	57.1	60.8	69.8	75.4	82.1	85.6	84.3	77.3	68.1	60.4	56.9		
		MCWB	51.9	51.8	51.5	57.2	61.9	65.9	66.8	66.1	63.0	61.1	56.9	53.3		
	2%	DB	52.2	53.6	57.3	64.1	71.3	75.6	80.4	78.7	71.7	64.1	57.3	55.0		
		MCWB	49.1	49.5	50.6	54.0	59.5	62.1	64.7	65.2	61.6	58.7	54.3	52.1		
	5%	DB	50.3	51.6	55.0	60.5	66.5	71.7	76.7	73.7	68.1	61.1	55.4	52.1		
		MCWB	47.7	47.9	49.5	52.2	57.1	60.6	63.4	62.7	59.8	57.0	52.2	49.4		
	10%	DB	48.4	49.7	52.1	57.2	62.9	68.2	72.9	71.2	65.8	59.3	53.4	50.1		
		MCWB	46.3	46.6	47.3	50.3	55.0	59.2	62.0	61.6	58.3	55.4	50.5	47.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	53.0	53.9	58.4	63.5	67.7	69.4	70.3	65.6	62.0	57.2	54.2		
		MCDB	55.0	56.1	58.4	67.2	71.8	78.8	79.1	80.1	72.9	66.5	59.8	56.4		
	2%	WB	50.2	50.4	51.9	55.2	60.3	63.9	66.6	66.3	63.1	59.6	55.0	52.5		
		MCDB	52.1	52.9	56.0	61.9	68.8	72.0	76.8	75.0	69.6	63.2	57.3	54.8		
	5%	WB	48.3	48.4	50.1	52.9	58.3	61.9	64.7	64.4	61.0	57.9	52.9	50.5		
		MCDB	50.2	51.1	53.6	59.3	65.5	69.5	73.5	71.6	66.2	61.1	55.3	52.4		
	10%	WB	46.4	46.5	48.3	50.9	56.1	60.1	63.0	62.7	59.5	56.0	51.0	48.1		
		MCDB	48.4	49.1	52.1	56.2	61.7	66.8	71.1	69.1	64.1	59.0	53.1	50.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.8	12.0	14.3	15.1	15.2	15.9	15.0	13.5	11.2	9.3	8.3		
		MCDBR	10.1	11.1	15.0	18.6	20.8	21.5	22.4	19.7	17.4	12.3	9.6	9.5		
		MCWBR	9.6	8.8	10.2	11.0	11.8	11.2	9.9	9.5	9.6	8.7	8.7	9.1		
	5% WB	MCDBR	10.1	10.5	13.4	17.2	18.9	18.9	19.6	16.8	14.7	11.7	9.6	9.7		
		MCWBR	10.0	9.2	10.1	11.0	11.4	11.3	10.7	9.4	9.8	9.1	9.0	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.338	0.370	0.396	0.400	0.401	0.400	0.387	0.368	0.350	0.324	0.311		
	taud		2.348	2.383	2.290	2.227	2.256	2.280	2.286	2.348	2.402	2.432	2.403	2.328		
	Ebn at Noon		208	244	257	264	269	270	268	265	256	236	207	188		
	Edh at Noon		19	25	33	40	40	40	39	35	29	23	18	16		
(44) All-Sky Solar Radiation	RadAvg		230	449	789	1213	1506	1592	1538	1269	945	547	284	187		
	RadStd		21	53	95	137	98	154	165	88	85	47	31	19		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (88 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page